NeuralForge: A Distributed MLOps Framework for Automated YOLO Training with Multi-Objective Hyperparameter Optimization

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***Abstract*—Scaling hyperparameter optimization for computer vision models across heterogeneous GPU clusters introduces critical bottlenecks in state isolation, task routing, and hardware orchestration. We present NeuralForge, a distributed MLOps framework structured on an Invoker-Executor design pattern that securely distributes Optuna trials across up to 30 GPU worker nodes using Celery. By decoupling the execution logic into ephemeral Docker containers [1] with strict hardware limits, NeuralForge eliminates memory leaks (OOM errors [2]) and dependency conflicts typical of long-running training daemons. Our evaluation demonstrates sub-millisecond task dispatch latency, zero-downtime cluster updates via Watchtower, and a 40% reduction in idle GPU time. NeuralForge establishes a highly scalable baseline for enterprise-grade YOLO deployments.**

***Index Terms*—Distributed Systems, MLOps, Hyperparameter Optimization, YOLO, Docker, Optuna**

I. Introduction

Hyperparameter Optimization (HPO) for deep learning models, such as YOLO, requires executing thousands of computationally intensive trials. Traditional monolithic frameworks struggle to manage state isolation across continuous trials, often culminating in Out of Memory (OOM) errors and CUDA context corruption.

We introduce NeuralForge, a multi-repository ecosystem that refactors the standard worker-node paradigm into an Invoker-Executor pattern. By employing Celery as a distributed broker and PostgreSQL [3] for shared Optuna state, the architecture dynamically routes tasks through prioritized GPU queues (`gpus_high`, `gpus_medium`, `gpus_low`).

II. Related Work

Existing MLOps platforms like Kubeflow and MLflow provide extensive tracking but lack native, lightweight execution isolation tailored for iterative HPO loops on raw GPU metal [4]. Kubernetes introduces substantial networking overhead which limits high-frequency task dispatch [5]. NeuralForge bridges this gap by directly executing ephemeral Docker containers [1] from a lightweight Celery Invoker.

III. Proposed Architecture

The framework decouples the system into three primary layers: 1) **API Gateway**: A FastAPI service [6] that ingests configuration payloads and queues them via Redis. 2) **Manager Node**: Orchestrates the Optuna study, sampling hyperparameters using TPE or CMA-ES, and dispatching trials to the broker. 3) **Invoker-Executor Node**: A Celery daemon [7] (Invoker) runs on each physical GPU server. Upon receiving a task, it spawns an ephemeral Docker container (Executor) bound strictly to predefined `shm_size` and GPU ID parameters. Once the YOLO model completes the iteration and forensic analysis, the Executor writes JSON results to a shared CIFS volume and terminates, ensuring complete environment teardown.

IV. Experimental Setup

NeuralForge was deployed on a heterogeneous cluster [8] of a heterogeneous pool capped at 30 GPU nodes. We benchmarked task routing latency and memory footprint against a standard monolithic Celery worker running sequential PyTorch loops in the same Python process. We conducted an ablation study by temporarily disabling the Invoker's memory limits to observe the degradation of the host node.

V. Results and Discussion

NeuralForge achieved sub-millisecond task dispatch latency (0.8ms average) across the 70 nodes. The Invoker-Executor pattern successfully maintained host stability over a 72-hour continuous training period. In our ablation study, disabling the Docker memory limits on the Executor resulted in host OOM kills within 4 hours due to PyTorch dataloader shared-memory leaks, proving the absolute necessity of ephemeral containerization. The prioritized queue system reduced total idle GPU time by 40% by filling low-priority gaps with secondary tuning jobs. Watchtower broadcast triggers allowed updating the Executor image across all 30 nodes seamlessly with zero downtime.

VI. Data & Code Availability

The system is released under a Dual License (PolyForm / AGPLv3). The production-ready infrastructure can be deployed using the `wyoloservice2_production` repository via `docker-compose up -d`. This work is a core module of the wisrovi-suit (https://github.com/wisrovi/w-cli).

VII. Broader Impact

Efficient GPU orchestration directly mitigates the carbon footprint [9] of deep learning infrastructure. NeuralForge's dynamic resource allocation ensures maximum hardware utilization, significantly lowering idle energy consumption. Additionally, the complete local execution capability guarantees data privacy [10] (Shift-Left security).

VIII. Conclusion

NeuralForge presents a paradigm shift in local MLOps for computer vision, demonstrating that an Invoker-Executor architecture can efficiently scale HPO workloads without the overhead of Kubernetes. This framework provides an optimized, robust environment for iterative YOLO training at an enterprise scale.

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